
بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

REMOTE SENSING SATELLTE ORBITS AND DATA ACQUISITION

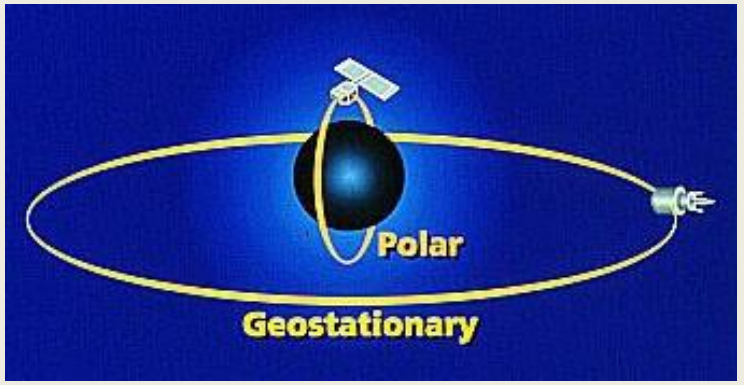


SATELLITE ORBITS

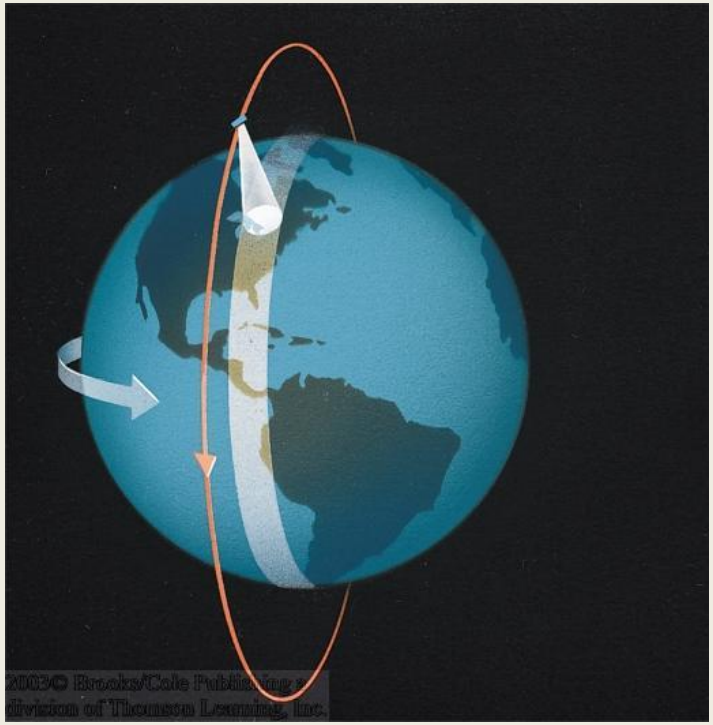
- Path followed by a satellite is referred to as its orbit
- The orbit into which a satellite is deployed is predetermined by what we require the satellite to observe
- Remote sensing satellites are designed such that their orbits match the capabilities of their sensors
- Orbit of RS satellites are chosen such that a large area is periodically under observation

SATELLITE ORBITS

- **GEOSTATIONARY**

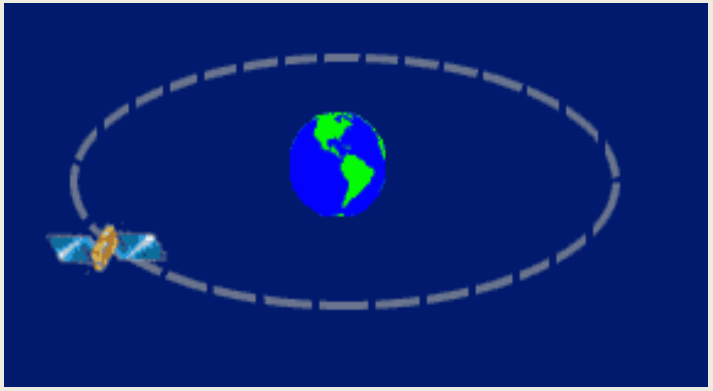


- **POLAR ORBITS**



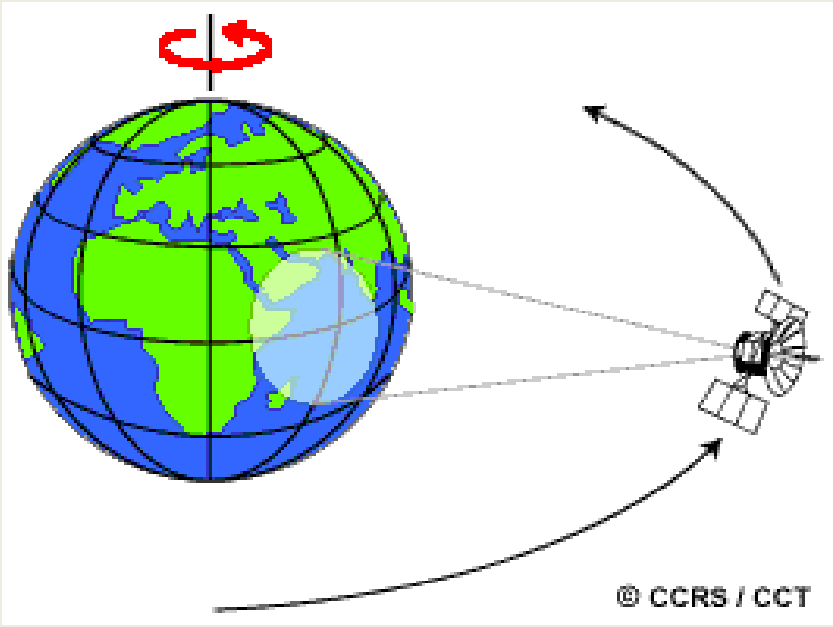
GEOSTATIONARY/ GEOSYNCHRONOUS ORBIT

- Stationary with reference to Earth
- Circle Earth at same rate as Earth spins
- Complete one revolution of the Earth in the same amount of time as of one rotation of the Earth about its axis, moving West to East
- Orbital altitude ~ 36000 km



GEOSTATIONARY ORBIT

- View the same portion of the Earth's surface at all times
- Almost full hemisphere observation
- Often used in meteorological and communication satellites



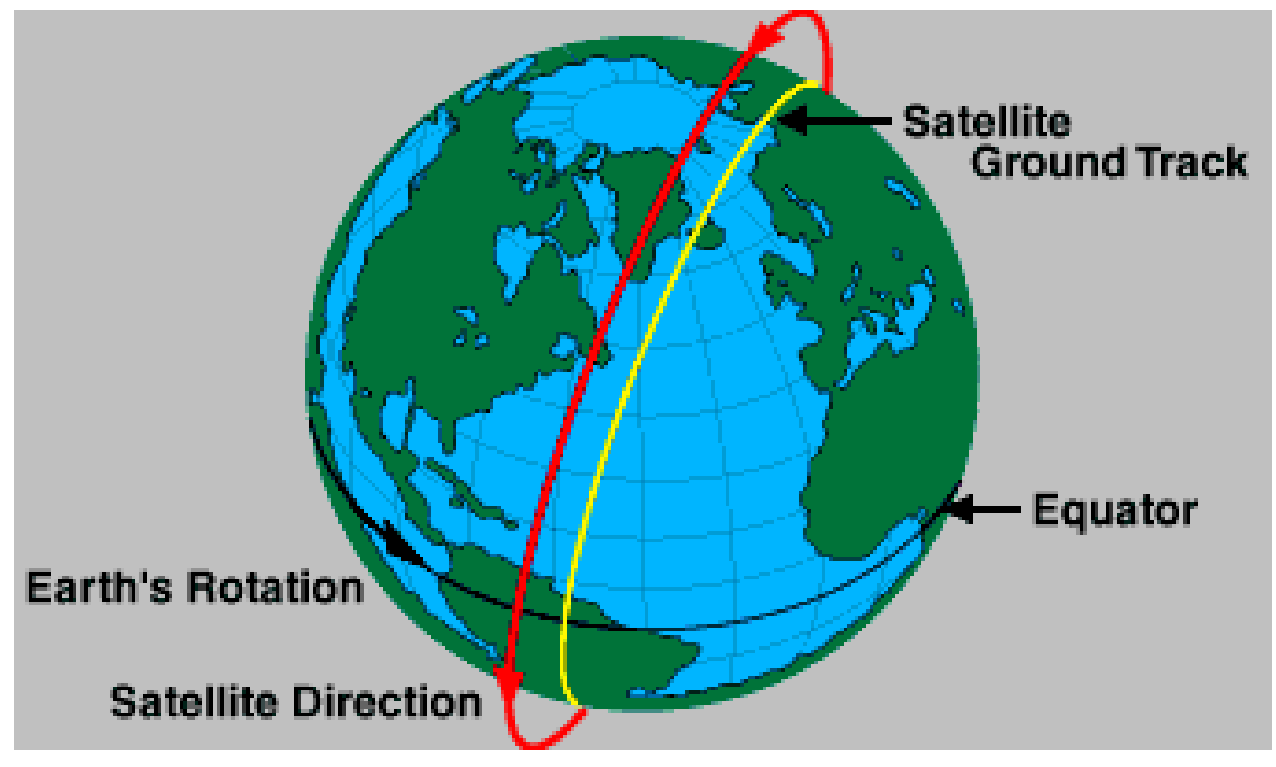
SATELLITE ORBITS

- **POLAR ORBITS**
  - **Orbiting the Earth at about 1000 km**
  - **Generally, in the North –South direction**



SUN-SYNCHRONOUS ORBIT
(near Polar)

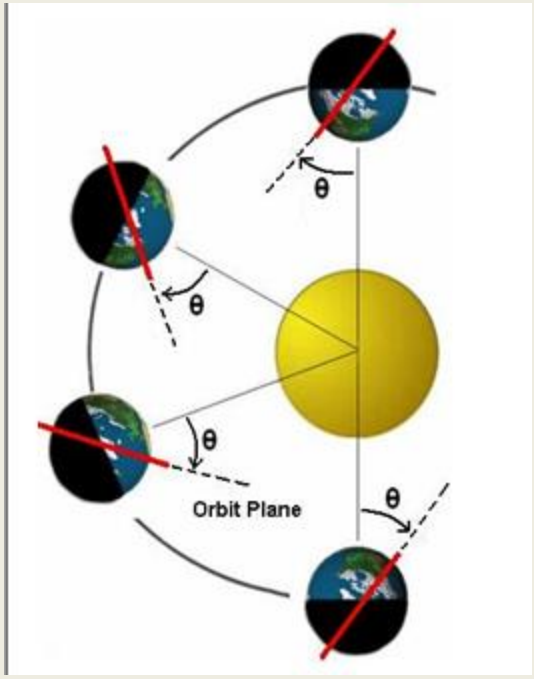
- **Satellite observe every point on the Earth at the same local time of the day (**local sun time)



SUN-SYNCHRONOUS ORBITS

- **Designed to ensure that the angle between the orbital plane and sun remains constant to ensure consistent illumination conditions**

- **Altitude <1000 km**
- **Θ = 37.5◦**



SATELLITE ORBITS

- Descending path (N to S) on the sunlit side of the earth and ascending on the shadowed side (S to N)
- Sensor that depend on reflected solar radiations are programmed to acquire data only during the descending path
- Time of day and seasons are responsible for the variations in the characteristics and intensity of light

SATELLITE ORBITS

- Earth observational satellites are usually placed in orbits designed to provide the best illumination and to minimize cloud cover
- One desirable effect of sun-synchronous orbit is that illumination is similar in images taken on different orbits

SATELLITE ORBITS

- **An early morning crossing time**
  - **The sun at a very low angle in the sky and would be good for emphasizing topographic effects but**
  - **Would result in a lot of shadow in areas of high relief**

SATELLITE ORBITS

- **A crossing time around noon**
  - **Would have the sun at its highest point in the sky and would provide the maximum and most uniform illumination conditions**
    - **This would be useful for surfaces of low reflectance but might cause saturation of the sensor over high reflectance surfaces, such as ice**
  - **Also, under such illumination, ‘mirror like reflection' from smooth surfaces may be a problem for interpreters**

SATELLITE ORBITS

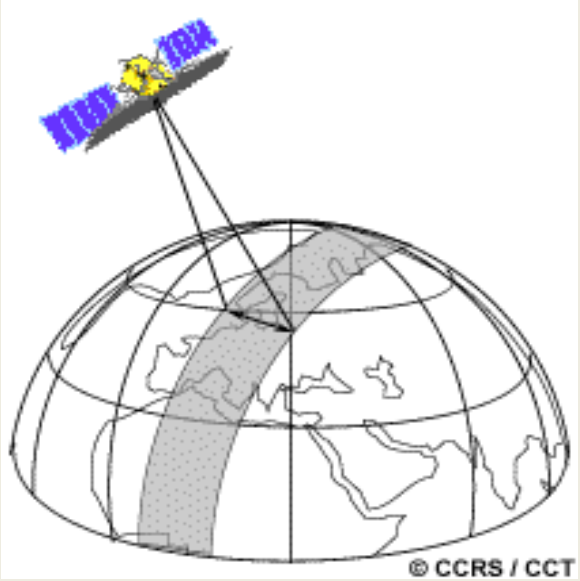
- **In the mid afternoon**
  - **The illumination conditions would be more moderate**
  - **However, a phenomenon called solar heating (due to the sun heating the surface), which causes difficulties for recording reflected energy, will be near maximum at this time of day**
- **To minimize these effects, most satellites that image in the visible, reflected, and emitted infrared regions use crossing times around mid-morning as a compromise**

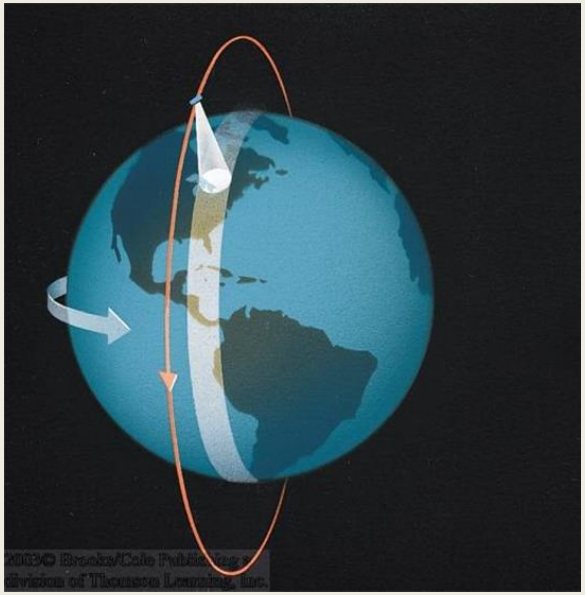
SATELLITE ORBITS

- **Equatorial crossing**
  - **The points where the path crosses the equator**
- **Orbital inclination**
  - **The angle between the satellite track and the equator, (Descending path)**
  - **Controls the north-to-south latitude extent of the satellite orbit**
- **Orbital Time/Period**
  - **The time a satellite takes to complete one orbit, the period increases with the altitude, for variable altitudes time will vary**
  - **At an altitude of 36000 km, sat has same period as the earth**
- **Altitude**

SATELLITE SWATH

- During the senor's movement, it sees certain area on the surface of earth ,is referred as swath
- Swath: Scanning width, the satellite sensors covers
- It helps determine the scene size



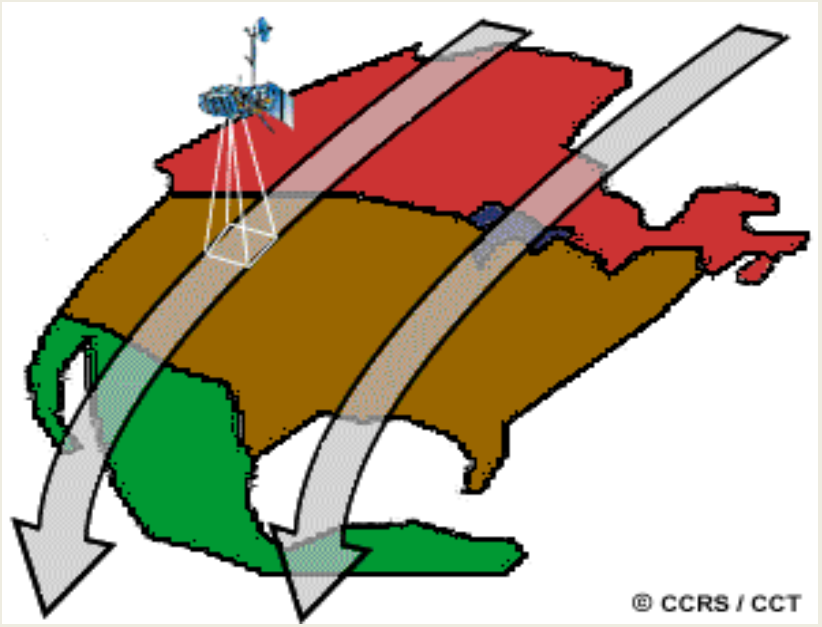


SWATH

- As the satellite orbits the Earth from pole to pole, its east west position wouldn't change if the earth didn't rotate
- From the orbit's pattern it seems that satellite is shifting westward because earth is rotating west to east beneath it
- This apparent movement allows the satellite swath to cover a new area with each consecutive pass
- Satellite's orbit and rotation of the earth work together to allow complete coverage of earth's surface after it has completed one complete cycle of orbit

ORBITS VS EARTH ROTATION





- **From one orbit to the next, the position directly below the sat moves a certain distance at the equator as the earth rotate beneath it**

ORBIT CYCLE/PERIOD/TIME

$$T = 2\pi(R+H)\sqrt{\frac{(R+H)}{gR^2}}$$

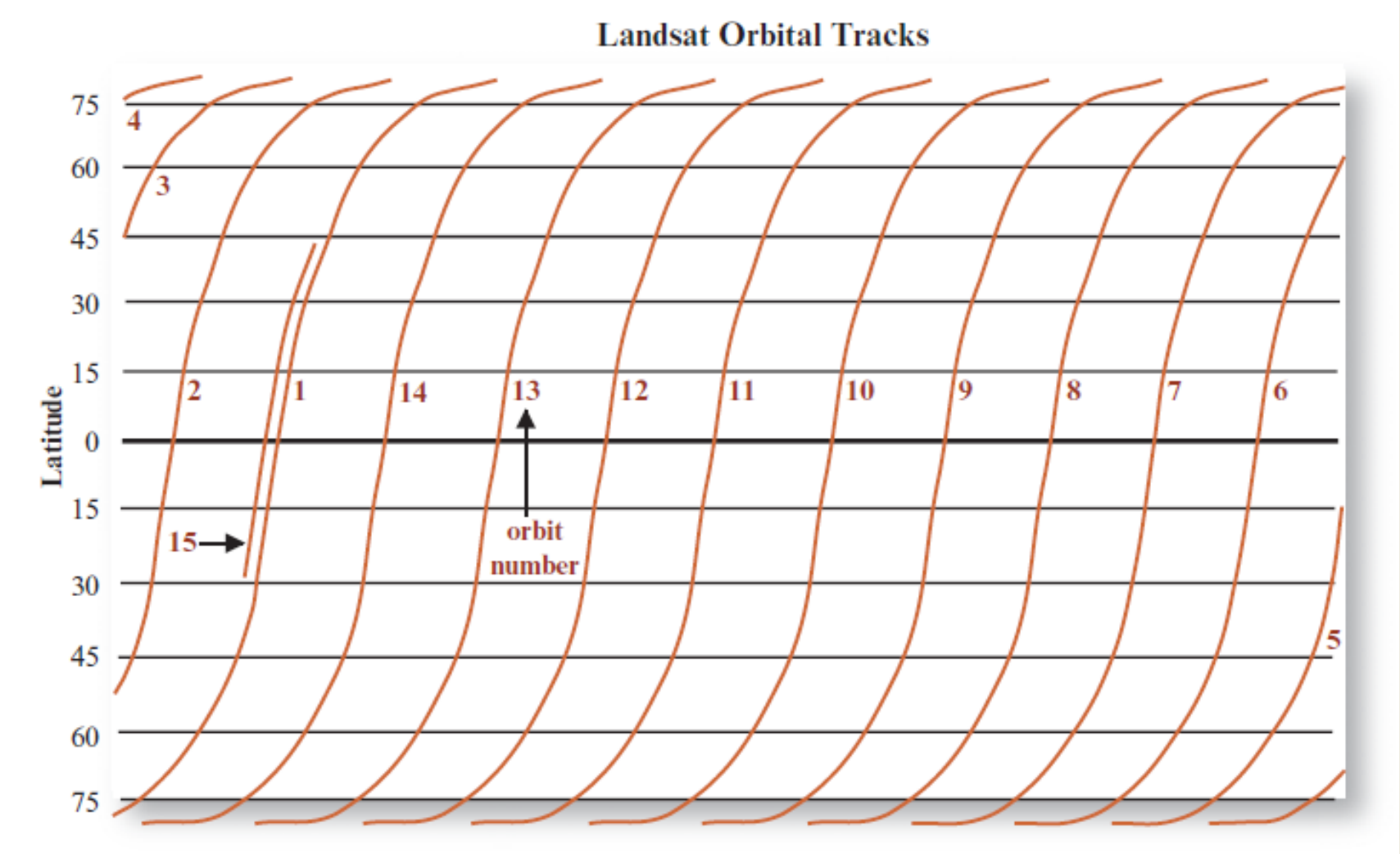
- T- Orbit period
- R- radius of the Earth ≈(6380 km)
- H-Orbital altitude=917 Km
- g- Acceleration due to gravity= 0.00918 km/sec

ORBIT CYCLE/PERIOD

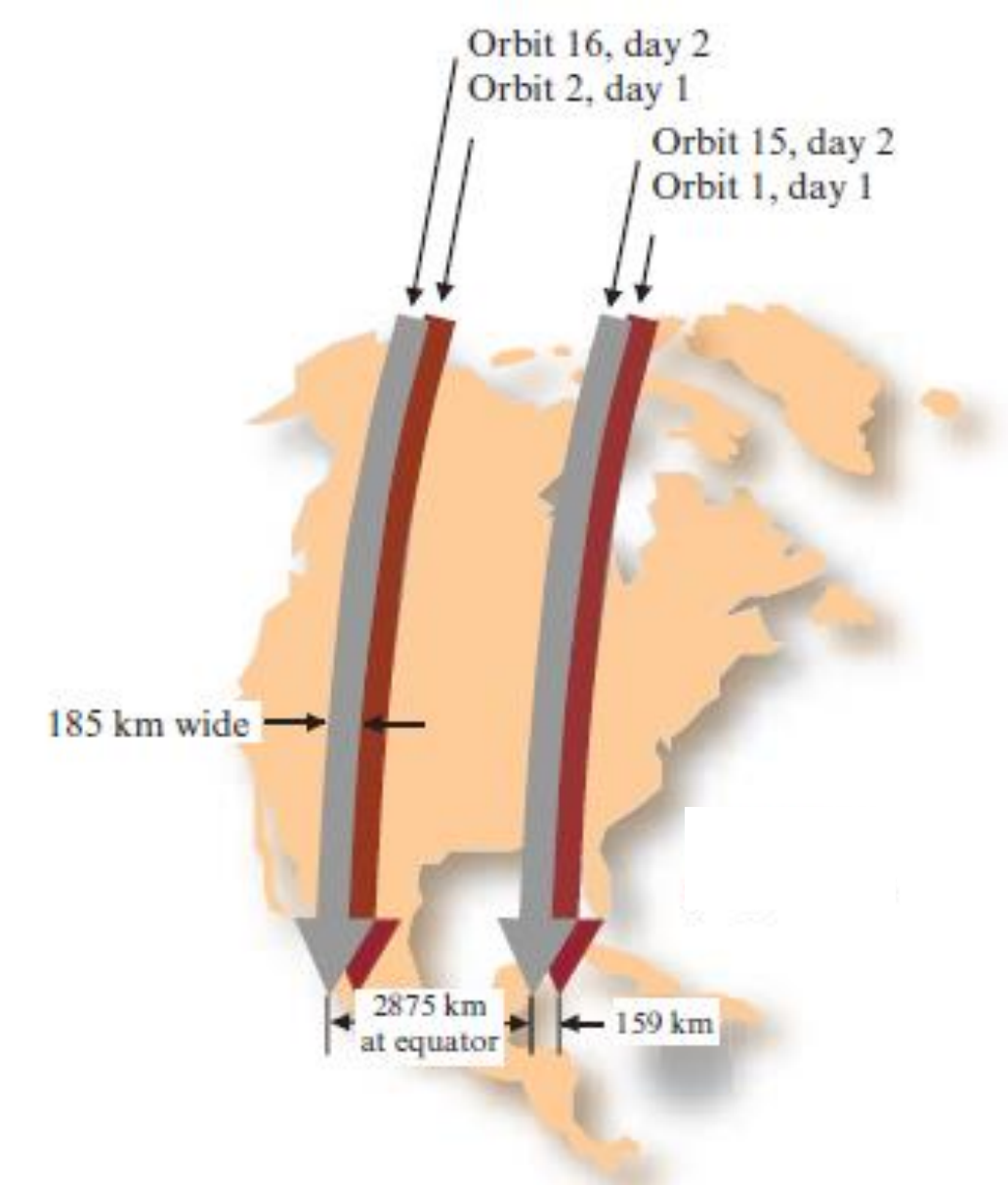
$$N = \frac{\text{Tearth}}{\text{Torbit}}$$

- T_{orbit} = Time to complete one orbit
- T_{earth} =Time Earth takes to complete its one orbit
- **N= Number of orbits per day**

LANDSAT COVERAGE



COVERAGE


Orbit 16, day 2
Orbit 2, day 1
Orbit 15, day 2
Orbit 1, day 1
185 km wide
2875 km
at equator
159 km

ORBITS OVERLAP

$$D = \frac{2\pi R}{T_{earth}} x T_{orbit}$$

D= distance Earth rotated between the two orbits

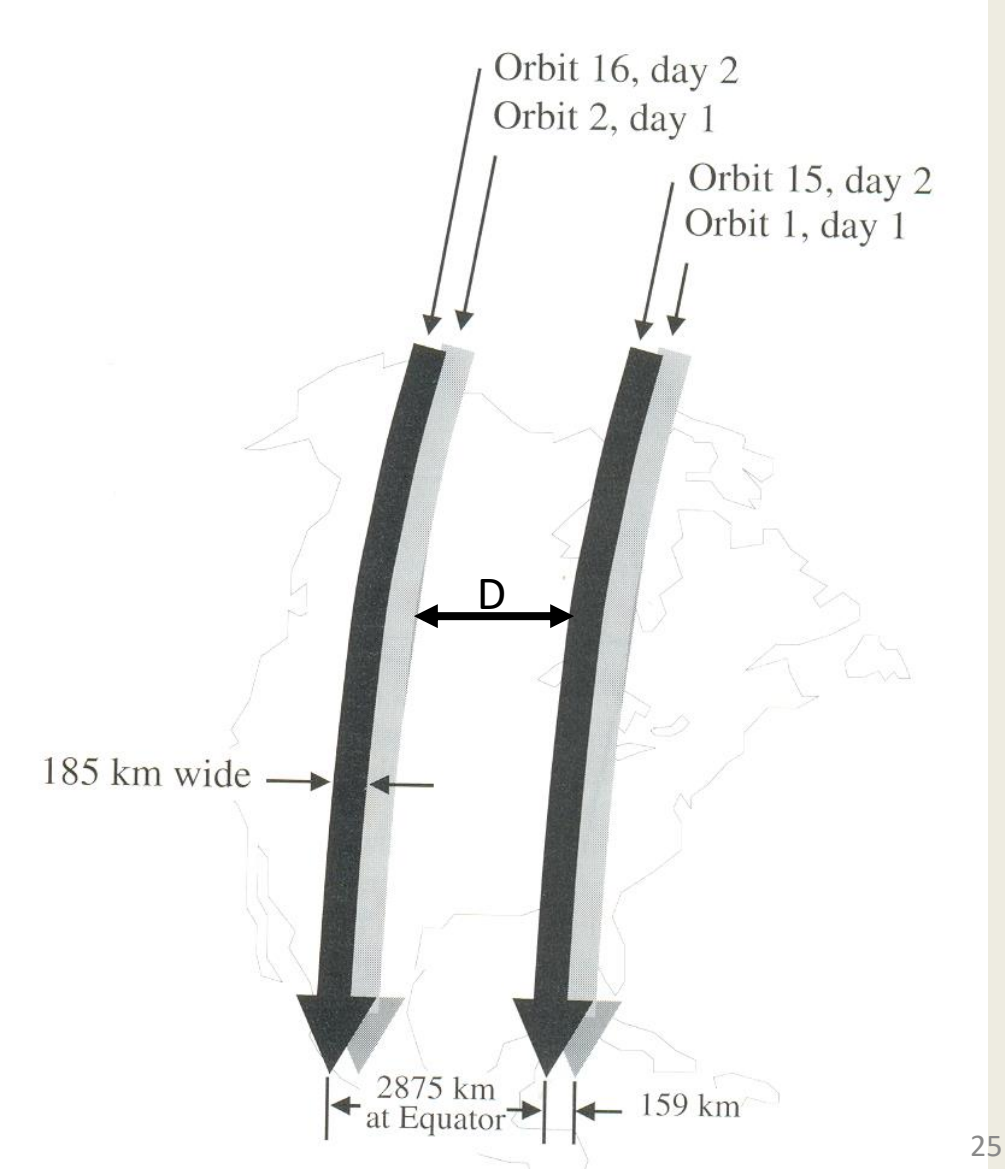
R= Radius of Earth

T_{orbit} = Time to complete one orbit

T_{earth} =Time earth takes to complete its one orbit

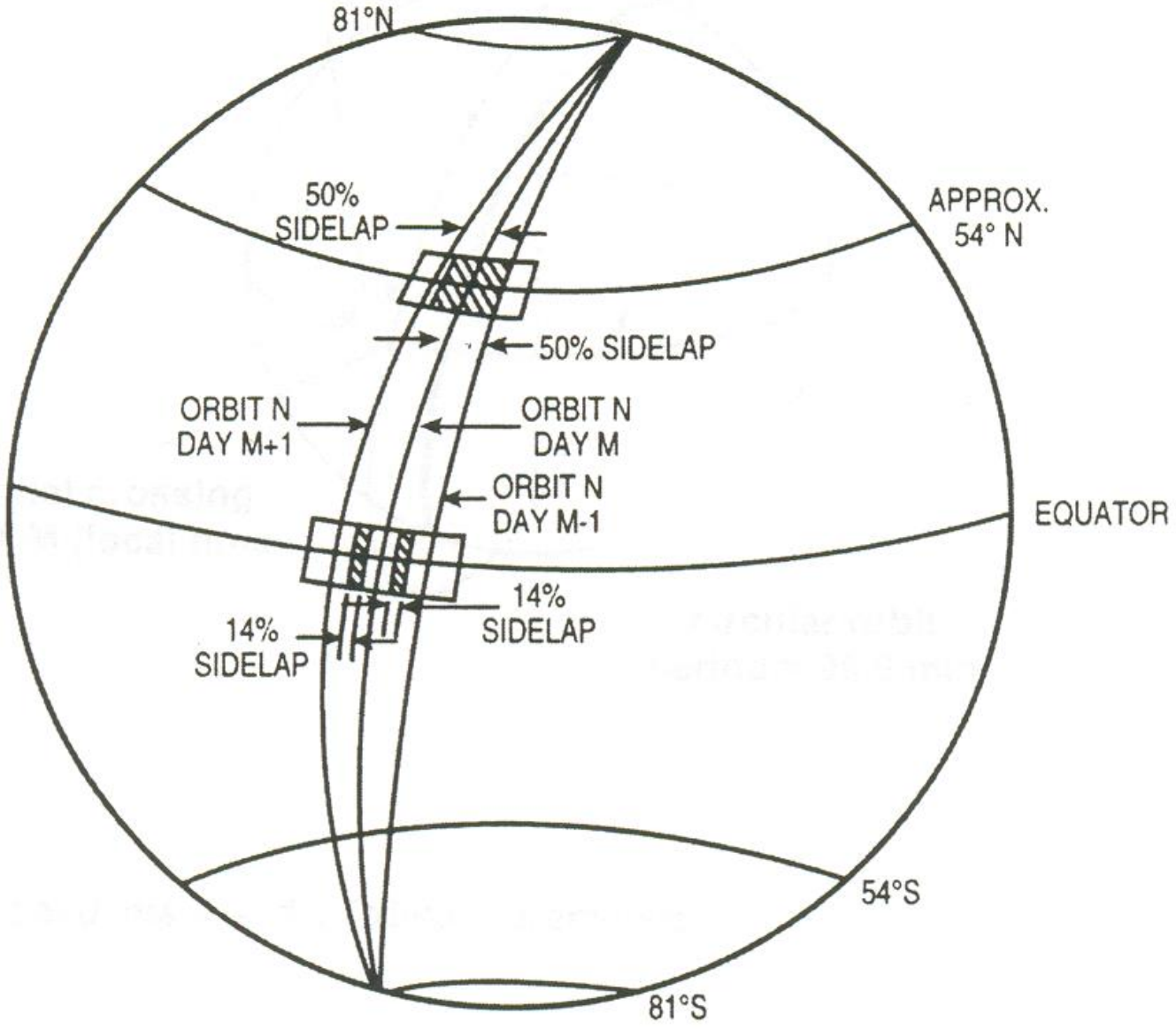
Westward longitudinal shift 159 Km

To observe the entire globe 2875/159= 18 days (20 times a year)



REVISIT PERIOD

- The time after which the satellite revisits the same portion od earth surface
- Revisit period is important for frequent imaging requirements
- In near polar orbits, areas at high latitudes will be imaged more frequently than the equatorial zone due to the increasing overlap in adjacent swaths as orbit paths come closer near the poles


81°N
50%
SIDELAP
APPROX.
54° N
50% SIDELAP
ORBIT N
DAY M+1
ORBIT N
DAY M
ORBIT N
DAY M-1
EQUATOR
14%
SIDELAP
14%
SIDELAP
54°S
81°S

